

The Modulus Map Refutes the Direct Non-Monotone Sublinear Wulbert Analogue in $L^1[0, 1]$

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Status

Candidate full counterexample to the direct analogue; likely valid, wording-scope review needed. The source asks informally for “an analogue.” The result below completely refutes the direct formulation obtained by retaining Wulbert’s hypotheses and replacing linear maps by continuous sublinear maps. A modified theorem with sign-symmetric tests is not claimed to be false.

Source question and precise direct reading

Gal–Niculescu ask on PDF page 12 of arXiv:2206.14102 whether Wulbert’s non-monotone Korovkin theorem for operators on $L^1[0, 1]$ admits an analogue for continuous sublinear maps.

$$\begin{aligned} \|T_n(J) - J\| &\leq \|T_n(J) - T_n(\mu)\|_{L^p} + \|T_n(\mu) - \mu\|_{L^p} + \|\mu - J\|_{L^p} \\ &\leq M \|f - h\|_{L^p} + \|T_n(h) - h\|_{L^p} + \|h - f\|_{L^p}, \end{aligned}$$

allows us to conclude that $\|T_n(f) - f\|_{L^p} \rightarrow 0$. □

It is conceivable that Theorem 5 still works in the case of an arbitrary finite Borel measure on $\mathbb{R}^N$, but at the moment we lack a valid argument.

In the setting of linear and continuous operators acting on $L^1[0, 1]$, Wulbert [30] has proved a Korovkin type theorem that avoids the hypothesis of monotonicity. It is an open question whether his result admits an analogue within the framework of continuous sublinear operators $T_n : L^1[0, 1] \rightarrow L^1[0, 1]$.

5. AN APPLICATION TO LEBESGUE DIFFERENTIATION THEOREM

The Lebesgue differentiation theorem is an important result in real analysis that

Figure 1: The complete source question, arXiv:2206.14102v1, PDF page 12.

Wulbert’s Theorem 9 is restated in Theorem 1.3 of arXiv:2403.03476. For linear $L_n : L^1[0, 1] \rightarrow L^1[0, 1]$, its hypotheses are

1. $L_n(1) \rightarrow 1$ in norm;
2. $L_n(x) \rightarrow x$ and $L_n(x^2) \rightarrow x^2$ weakly;
3. $\|L_n\| \rightarrow 1$;

and the conclusion is $\|L_n f - f\|_1 \rightarrow 0$ for every $f \in L^1[0, 1]$. The direct sublinear analogue changes only the word “linear.”

Counterexample

Theorem 1 (Positive-test no-go). *Let $\mathcal{K} \subset L^1[0, 1]_+$ be any family. There is a constant sequence of continuous sublinear operators $T_n : L^1[0, 1] \rightarrow L^1[0, 1]$ such that $\|T_n\| = 1$, $T_n h = h$ for every $h \in \mathcal{K}$, but T_n does not converge to the identity even weakly on all of $L^1[0, 1]$.*

Proof. For every n , set

$$T_n f = |f|.$$

The pointwise triangle inequality and positive homogeneity give

$$T_n(f + g) \leq T_n f + T_n g, \quad T_n(\alpha f) = \alpha T_n f \quad (\alpha \geq 0),$$

which is exactly the source's definition of sublinearity. Also

$$\|T_n f - T_n g\|_1 = \||f| - |g|\|_1 \leq \|f - g\|_1, \quad \|T_n f\|_1 = \|f\|_1.$$

Thus T_n is continuous (indeed 1-Lipschitz) and its nonlinear operator norm is one. Every $h \geq 0$ is fixed exactly.

On the other hand,

$$T_n(-1) = 1, \quad \|T_n(-1) - (-1)\|_1 = 2.$$

Pairing the difference with the constant member 1 of $L^\infty[0, 1]$ also gives 2, so even weak convergence fails. $\square$

Taking $\mathcal{K} = \{1, x, x^2\}$ verifies all three Wulbert hypotheses in strictly stronger form: each test is fixed in norm and $\|T_n\| = 1$ exactly. The conclusion nevertheless fails at -1 . The map is admissibly nonmonotone, since $-1 \leq 0$ but $T_n(-1) = 1 > T_n(0) = 0$.

Why this is the sharp literal obstruction

The counterexample does not exploit a badly behaved or discontinuous map: it is an isometry in norm and a 1-Lipschitz lattice operation. More importantly, it fixes the entire positive cone. Consequently no finite or infinite test family consisting only of nonnegative functions can yield identity convergence for all continuous sublinear contractions. Wulbert's linearity silently supplies oddness, turning information about positive tests into information about their negatives; plain sublinearity does not.

A viable repair must therefore add sign information or impose asymptotic oddness. Indeed, sublinearity always gives

$$0 \leq T_n(h) + T_n(-h).$$

If both terms converge weakly to h and $-h$, respectively, this nonnegative defect converges to zero in L^1 norm after testing against the constant one. This eliminates the modulus map on tested directions, but it does not by itself recover Wulbert's propagation from the test set to all of L^1 .

Upgrade and novelty audit

Six focused routes were checked: the literal modulus construction, arbitrary positive test families, the sign-symmetric defect, maxima of linear contractions, scaled/translated corrections, and later-literature searches. Attempts to preserve both signs with a nonidentity norm-one maximum map

run into Wulbert rigidity and the loss of the L^1 norm bound. No stronger sign-symmetric proof or counterexample is promoted.

The exact source sentence and combinations of Wulbert, continuous sublinear, L^1 , non-monotone, and modulus were searched in the local corpus, arXiv, and bounded citation neighborhoods through 2026-08-11. The later arXiv:2403.03476 generalizes Wulbert within linear operators and confirms the three hypotheses, but does not treat sublinear maps. No indexed prior use of this obstruction for the 2022 question was found; novelty remains subject to specialist review.

References

- [1] S. G. Gal and C. P. Niculescu, *Korovkin type theorems for weakly nonlinear and monotone operators*, arXiv:2206.14102, PDF page 12.
- [2] D. E. Wulbert, *Convergence of operators and Korovkin's theorem*, J. Approx. Theory 1 (1968), 381–390, Theorem 9.
- [3] V. B. Kiran Kumar, M. N. N. Namboodiri, and P. C. Vinaya, *Korovkin-type approximation for non-positive operators*, arXiv:2403.03476v2, Theorem 1.3.